

Answer Key — Biology Grade 11

Sub-Strand 1.4: Cell Division • Lesson 7: Microscope Practical on Mitosis: Catching Growth in Action in Onion Root Tips

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
A	A	A	B	B	A	D	D

USE AFTER: PREDICT PHASE — Predict: onion root grown in a bottle of water, prediction sketch and Think-Pair-Share

Q1. A onion (kitunguu) root grown in water for a week lengthened by 3 cm. Which part of the root is making most of the new cells that make it longer?

- A. The tip region, where cells are dividing**
- B. Nowhere, because the existing cells just get bigger
- C. The middle only, where the root is oldest
- D. The whole root equally, from the tip to the top

Correct answer: A. The root lengthens because cells near the tip divide to make new cells. The other options reflect the misconceptions that the whole root grows equally or that cells simply get bigger.

USE AFTER: OBSERVE PHASE — Observe: preparing the onion root tip squash (steps 1 to 6)

Q2. In preparing an onion root tip squash, the tip is warmed in 1M hydrochloric acid for a few minutes. Why is this done?

- A. To soften the tissue so the cells can be squashed into a thin single layer**
- B. To colour the chromosomes
- C. To make the cells divide faster
- D. To make each cell grow bigger

Correct answer: A. The acid softens the tissue so that when the slide is squashed the cells spread into a single layer that light can pass through. The stain (acetic orcein) is what colours the chromosomes.

USE AFTER: OBSERVE PHASE — Observe: drawing cells and identifying stages under the microscope (oral question "Which cell is in metaphase?")

Q3. Under the microscope, a learner sees a cell in an onion root tip where the dark-stained chromosomes are all lined up across the middle (equator) of the cell. Which stage of mitosis is this?

- A. Metaphase**
- B. Telophase
- C. Prophase
- D. Anaphase

Correct answer: A. Chromosomes lining up at the equator of the cell is the key feature of metaphase. In anaphase they have already been pulled apart.

USE AFTER: OBSERVE PHASE — Observe: tallying 50 to 100 cells by stage and calculating the proportion in each stage

Q4. A group counts 80 cells from an onion root tip squash. Of these, 60 are in interphase and 20 are in the stages of mitosis (prophase, metaphase, anaphase or telophase). What percentage of the cells are in interphase?

- A. 20%
- B. 75%**
- C. 25%
- D. 60%

Correct answer: B. Proportion in interphase = $60 \div 80 \times 100 = 75\%$. Most cells are in interphase, so mitosis is short compared with the whole cell cycle.

USE AFTER: EXPLAIN PHASE — Explain: Jigsaw, expert groups agreeing on a clue to identify each stage

Q5. A learner is looking at two cells in an onion root tip. What clue tells you that a cell is in anaphase and not metaphase?

- A. Chromosomes are lined up along the middle of the cell
- B. Sister chromatids are being pulled apart towards opposite ends of the cell**
- C. Two separate new nuclei have formed and the cell is dividing in two
- D. Chromosomes are thin and thread-like inside an intact nucleus

Correct answer: B. In anaphase, sister chromatids separate and move to opposite poles. Lined-up chromosomes show metaphase, and two new nuclei show telophase.

USE AFTER: EXPLAIN PHASE — Explain: Discussion Diamond ranking of the four statements using tally results

Q6. Learners see many cells at different stages of mitosis in an onion root tip, while cells further from the tip are not dividing. Which statement is best supported by this evidence?

- A. New cells are made when existing cells in the tip divide, and each new cell receives a copy of the chromosomes**
- B. The root grows only because its cells get bigger
- C. Most cells in the tip are dividing at exactly the same moment
- D. Food turns directly into new cells at the tip

Correct answer: A. Seeing chromosomes behave in distinct stages in the growing tip shows that new cells come from the division of existing cells. The tally also shows most cells are in interphase, so they are not all dividing at once.

USE AFTER: MODEL BUILDING PHASE — Model: revising the growth model, with the calculation 'If one cell divides every 24 hours, how many cells after 10 days?'

Q7. One cell divides by mitosis to make two cells, and every cell divides again every 24 hours. Starting with 1 cell, how many cells will there be after 10 days?

- A. 100
- B. 512
- C. 20
- D. 1024**

Correct answer: D. The number doubles each day (1, 2, 4, 8 ...), so after 10 days there are $2^{10} = 1024$ cells. This shows how repeated division can give billions of cells.

USE AFTER: END OF LESSON — Model: linking to growth, 'why a Kenyan child who eats well grows more than one who does not', and the exit ticket

Q8. A Kenyan child who eats well grows more than a child who does not get enough food. Which explanation fits what we learned about mitosis?

- A. Food makes cells divide without needing chromosomes
- B. Food turns directly into new cells in the body
- C. Food makes each cell grow bigger so no cell division is needed
- D. Food gives the materials and energy that cells need to divide and make new cells**

Correct answer: D. New cells are made by division of existing cells, and food supplies the materials and energy for this. Food does not turn directly into cells.